Bayesian SAE with an Emphasis on Low- and Middle-Income Countries: Area Level SAE Models

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Outline

Area-Level Modeling

Beyond Weighted Estimates

- **Aim of Small Area Estimation (SAE):** Provide inferential summaries across geographical areas, using survey data – in a LMICs context, the household surveys use stratified two-stage unequal probability cluster sampling.
- Recall the weighted estimator, which provides a simple approach and has many desirable properties in data-rich situations.
- As data become sparser, weighted estimates can be highly unstable (high variance) or we can't calculate a measure of uncertainty.
- One then needs to consider SAE models, of which there are two distinct types:

1: Area-Level Models	2: Unit-Level Models

Area-Level Models

The Area-Level Fay-Herriot Model

- An amazingly advanced model was introduced by Fay and Herriot (1979).
- The basic idea behind the area-level Fay-Herriot model, is to simultaneously model the collection of weighted estimates

$$\widehat{\theta}_i^{\mathrm{w}} = \frac{\widehat{T}_i}{\widehat{N}_i} = \frac{\sum_{k \in S_i} w_{ik} Y_{ik}}{\sum_{k \in S_i} w_{ik}}. \quad (1)$$

  from all areas.
- These estimates, $\widehat{\theta}_i^{\mathrm{w}}$, along with their standard errors, $\sqrt{V}_i$, constitute the data for $i = 1, \ldots, n$, areas.
- We model between-area variation in $\widehat{\theta}_i^{\mathrm{w}}$ using covariates and by assuming the residuals arise from a common distribution.
- We are effectively increasing the sample size in each area, by leveraging similarity in the means.

The most Basic Form of Area-Level Model

- We let $Y_i = \widehat{\theta}_i^{\text{W}}$ correspond to the "data" and $\text{E}[Y_i] = \theta_i$.
- The expectation is over samples that could have been drawn, and θ_i is the true finite population mean in area i.
- We describe a two-stage model in which we have a sampling model and a linking model.
- If there are sufficient samples within the area, a design-based asymptotic argument gives $Y_i \mid \theta_i \sim \text{N}(\theta_i, V_i)$.
- When we fit this model, it is an example of indirect estimation because the data from all areas is being used to enhance the estimates in each area.

The most Basic Form of Area-Level Model

The simplest version of the area-level model is:

$$
\begin{aligned}
Y_i &= \theta_i + \epsilon_i, \qquad \epsilon_i \sim_{ind} \mathrm{N}(0, V_i) && (2)\\
\theta_i &= \alpha + u_i, \qquad u_i \sim_{iid} \mathrm{N}(0, \sigma_u^2). && (3)
\end{aligned}
$$

- In the first stage of the model, in line (2) we have the sampling model with known variances V_i.
- The notation ind is short for independent.
- In the second stage of the model, in line (3) we have the linking model which links together $\theta_1, \ldots, \theta_n$.
- The notation iid is short for independent and identically distributed.
- Parameters: $\alpha = \mathrm{E}[\theta_i]$ is the intercept and is the average of the population means, $\sigma_u^2 = \mathrm{var}(\theta_i)$ is the variance of the residuals u_i.

Estimation for the Area-Level Model

- Suppose for simplicity that α and σ_u^2 are known.

- In this case the posterior mean estimate of θ_i is the so-called Best Linear Unbiased Predictor (BLUP), which is the posterior mean,

$$\begin{aligned} \widehat{\theta}_i^{\text{BLUP}} = \mathrm{E}[\theta_i \mid y_i] &= Y_i - \frac{V_i}{V_i + \sigma_u^2}(Y_i - \alpha) \\ &= (1 - q_i) \times Y_i + q_i \times \alpha \end{aligned}$$

  where

$$q_i = \frac{V_i}{V_i + \sigma_u^2} \leq 1$$

  is the shrinkage factor – estimates are shrunk towards α.

- A measure of uncertainty is the posterior variance $\mathrm{var}(\theta_i \mid y_i) = (1 - q_i) \times V_i$, so that the uncertainty is reduced.

- Notice that if σ_u^2 is big and/or V_i is small, we have $q_i \approx 0$ so that $\widehat{\theta}_i^{\text{BLUP}} \approx Y_i$ and $\mathrm{var}(\theta_i \mid y_i) \approx V_i$.

Fay-Herriot Model with Covariates

To model between-area variation in Y_i we may introduce area-level covariates $\boldsymbol{x}_i$, a $p \times 1$ vector.

With covariates:

$$\begin{aligned} Y_i &= \theta_i + \epsilon_i, \qquad \epsilon_i \sim_{ind} \mathrm{N}(0, V_i) & (4) \\ \theta_i &= \alpha + \boldsymbol{x}_i^{\intercal}\boldsymbol{\beta} + u_i, \qquad u_i \sim_{iid} \mathrm{N}(0, \sigma_u^2). & (5) \end{aligned}$$

- The BLUP becomes:

$$\begin{aligned} \widehat{\theta}_i^{\mathrm{BLUP}} = \mathrm{E}[\theta_i \mid y_i] &= Y_i - \frac{V_i}{V_i + \sigma_u^2}(Y_i - \alpha - \boldsymbol{x}_i^{\intercal}\boldsymbol{\beta}) \\ &= (1 - q_i) \times Y_i + q_i \times (\alpha + \boldsymbol{x}_i^{\intercal}\boldsymbol{\beta}) \end{aligned}$$

with $q_i = V_i/(V_i + \sigma_u^2)$, as before.

Including Covariates

- Covariates derived from a census are desirable but in LMICs censuses are infrequently carried out, and the data may not available at the level required.
- Instead, satellite data are often used.
- It is appealing to use survey covariates, but there are two issues that require consideration.

Survey-Based Covariates

- For simplicity assume a univariate x_i – if the same response data Y_i and covariate data X_i (the version with error) are from the same survey then we violate a key assumption of the model.
- We have

$$\begin{aligned} Y_i &= \theta_i + \epsilon_i \\ X_i &= x_i + \eta_i \end{aligned}$$

- Because both the area-level response Y_i and the auxiliary covariate X_i are direct (weighted) estimates derived from the same survey sample, the standard Fay-Herriot assumption of mutually independent sampling errors $\text{cov}(\epsilon_i, \eta_i) = 0$ is violated.
- The shared sampling variance induces an endogeneity problem via area-level measurement error.
- A separate issue, which is common to all covariates measured with error, is how to account for measuring X_i rather than x_i.

The Linear Mixed Effects Model

We can write the two-stage model (4) and (5) as

$$Y_i = \alpha + \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_i + \epsilon_i,$$

with

$$u_i \sim_{iid} \mathrm{N}(0, \sigma_u^2), \qquad \epsilon_i \sim_{ind} \mathrm{N}(0, V_i), \qquad i = 1, \ldots, n.$$

In this model:

- α and $\boldsymbol{\beta}$ are fixed effects.
- $u_1, \ldots, u_n$ are random effects – they have a distribution!
- Hence, we have a mixed effects model, and it is linear, and so, more specifically, it is a linear mixed effects model (LMEM).
- Note: This is a frequentist distinction since in a Bayesian framework, all parameters are random.

A Spatial Area-Level Model

Neighborhood Scheme in Rwanda

- Neighbors are conventionally defined through sharing a **common boundary** – this is a common choice but others are possible.

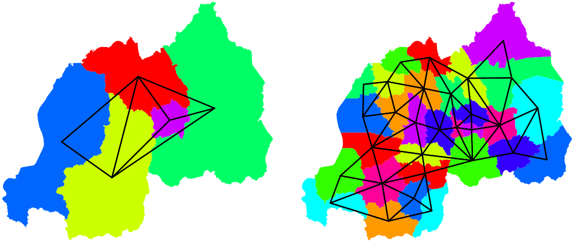

Figure 1: Neighborhood structure for 5 Admin1 areas and 30 Admin2 areas in Rwanda – this is used in the spatial smoothing models. Black lines link neighbors.

Area-Level (Fay-Herriot) Model

A wrinkle is that prevalences lie between 0 and 1, which makes modeling more tricky, so we take the logit transform.

Let,

$$\theta_i = \log\left(\frac{p_i}{1-p_i}\right),$$

be the log odds of the observed prevalence, which is on the real line.

The response variable is taken as,

$$Y_i = \log\left(\frac{\widehat{p}_i^{\text{w}}}{1-\widehat{p}_i^{\text{w}}}\right), \quad \text{for } i = 1, \dots, n, \text{ areas}$$

Data:

$$\underbrace{(Y_1, \sqrt{V}_1), (Y_2, \sqrt{V}_2), \quad \dots, \quad ,(Y_{n-1}, \sqrt{V}_{n-1}), (Y_n, \sqrt{V}_n)}_{\text{AREAS ARE LINKED THROUGH NEIGHBORHOOD STRUCTURE}}$$

Area-Level (Fay-Herriot) Model

In large samples,

$$Y_i = \underbrace{\theta_i}_{\text{TRUTH}} + \underbrace{\epsilon_i}_{\text{SAMPLING ERROR}},$$

with

- ϵ_i following a N(0, V_i) distribution,
- V_i estimated using formulas relevant to the survey design.

In the Fay-Herriot spatial model, it is assumed that

$$\underbrace{\theta_i}_{\text{TRUTH}} = \underbrace{\alpha}_{\text{OVERALL LEVEL}} + \underbrace{\delta_i}_{\text{SPATIAL RESIDUAL}}$$

The spatial terms δ_i are assumed to have spatial structure, i.e., they are spatially linked, strength of smoothness is controlled with a tuning parameter that is estimated from the data.

The spatial model we use corresponds to the celebrated BYM2 model (Riebler *et al.*, 2016), with the name based on the original Besag, York, Mollié model (Besag *et al.*, 1991).

Area-Level (Fay-Herriot) Model

The spatial residual is decomposed as:

$$\boldsymbol{\delta} = \begin{bmatrix} \delta_1 \\ \vdots \\ \delta_n \end{bmatrix} = \lambda \left(\sqrt{\phi} \begin{bmatrix} S_1 \\ \vdots \\ S_n \end{bmatrix} + \sqrt{1-\phi} \begin{bmatrix} E_1 \\ \vdots \\ E_n \end{bmatrix} \right)$$

where

- S_i is the spatial term and $E_i \sim \text{N}(0,1)$ is the non-spatial term,
- Two parameters in the model: ϕ is the proportion of the variation that is spatial and λ is the residual standard deviation – tells us how bumpy the surface is and corresponds to the smoothing parameter.
- For the spatial terms:

  $$S_i \mid S_j, \; j \in \text{ne}(i) \quad \text{follow a} \quad \text{N}(\overline{S}_i, 1/m_i) \quad \text{distribution,}$$

  where $\text{ne}(i)$ is the set of neighbors of i, $\overline{S}_i$ is the mean of the neighbors and m_i is the number of such neighbors – a local smoothing model.

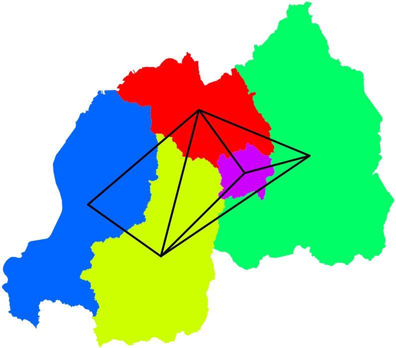

FIGURE 2: Neighborhood structure for 5 Admin1 areas. Black lines link neighbors.

- For example, the level for area 1 (note: numbering is arbitrary) is being pulled towards the average of the levels in areas 2, 3, 4 (its neighbors):

$$\mathrm{E}[S_1 | S_2, S_3, S_4] = \frac{1}{3}(S_2 + S_3 + S_4)$$

Advantages:

- Weighted estimates and uncertainty measures that feed into the hierarchical model account for the design, which avoids potential bias due to (say) stratified and PPS sampling, and accounts for the clustering.
- As the sample size in each area increases, the estimates tend to the true prevalences, which is known as design consistency.
- By modeling all the data in unison, uncertainty in each area is reduced (on average).

Disadvantages:

- The modeling is based on the weighted estimates and standard errors in each area, so when the data are sparse and estimates are not available or unreliable, the method cannot be used.
- When the data are sparse, the estimates may be overshrunk since the data are not sufficiently informative to discriminate from the “flat” map case.
- The model produces shrinkage which can lead to interpretation being more tricky (since bias is introduced in each estimate).

Subnational NMR Estimation in Rwanda: Area-Level Models

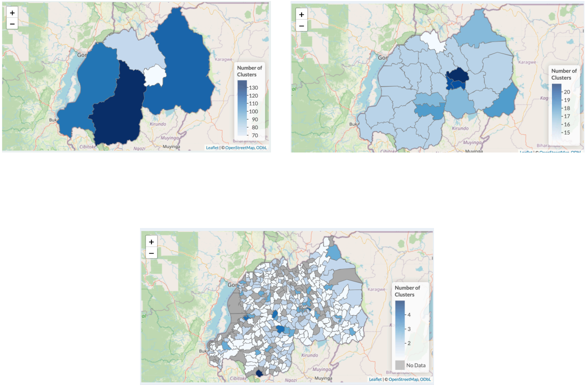

Figure 3: Number of sampled clusters, for Admin1/Admin2/Admin3 areas, from Rwanda 2019 DHS.

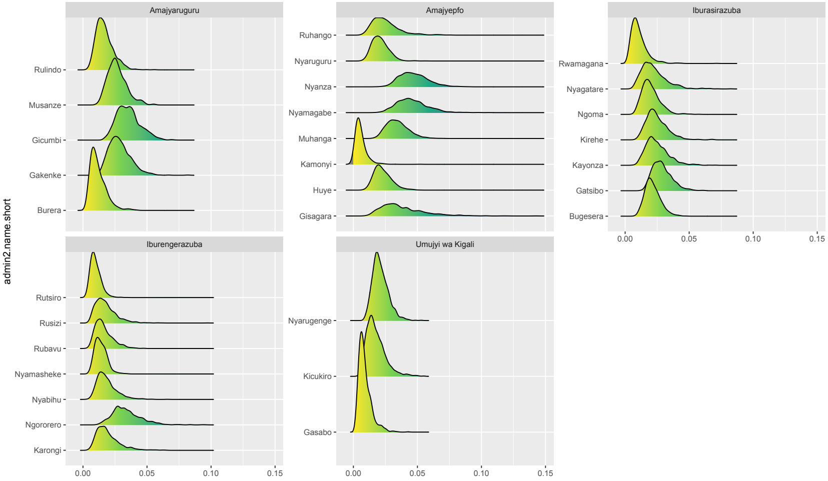

FIGURE 4: Ridgeplots for Admin2 areas *within* Admin1, direct estimates.

- For some Admin1, the constituent Admin2 estimates have similar distributions, while for others they are quite different.

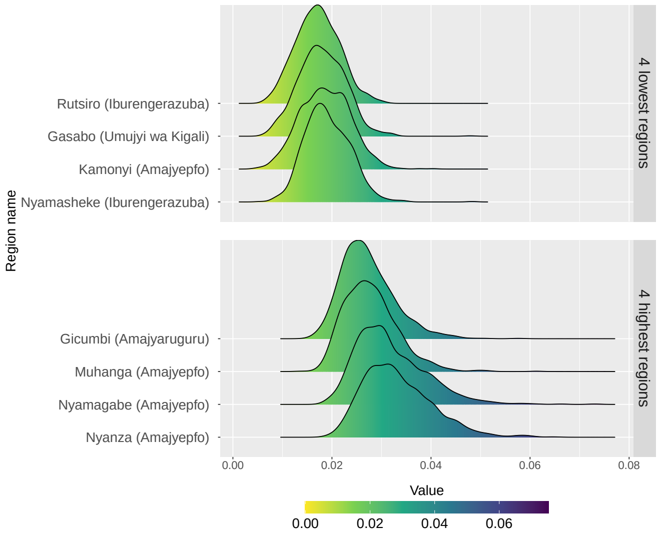

Figure 5: Ridgeplots for extreme (low and high) Admin2 areas.

- Lack of information at Admin2, which explains the relatively wide distributions and large overlap across areas.

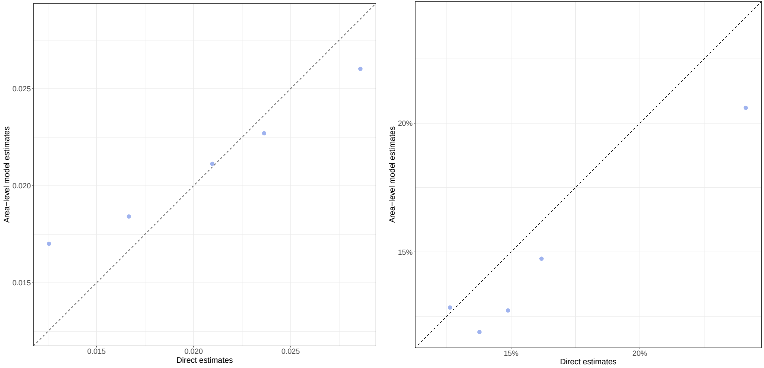

FIGURE 6: Area-level estimates versus direct estimates (left) and area-level CVs versus direct CVs (right), at Admin1.

- Some shrinkage (narrowing of range of estimates) of estimates at Admin1 (left).
- Gains in precision in all but one Admin1 area (right).

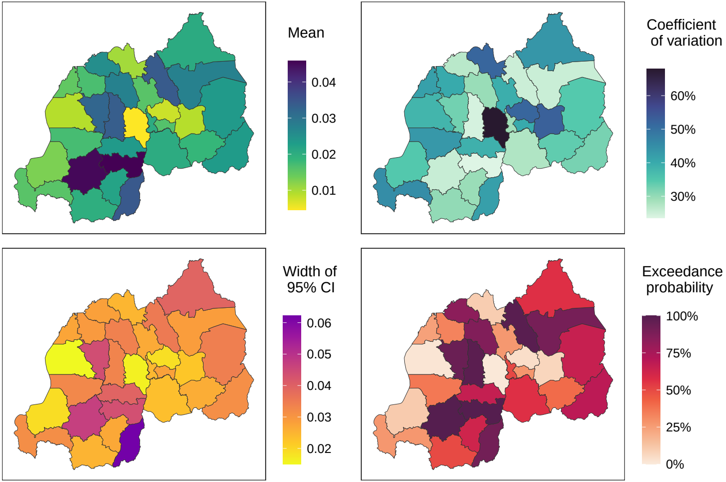

FIGURE 7: Summary measures for direct estimates at Admin2.

- Relatively large range in point estimates.
- Uncertainty measure show large variation also, and for some areas, estimates are quite imprecise.

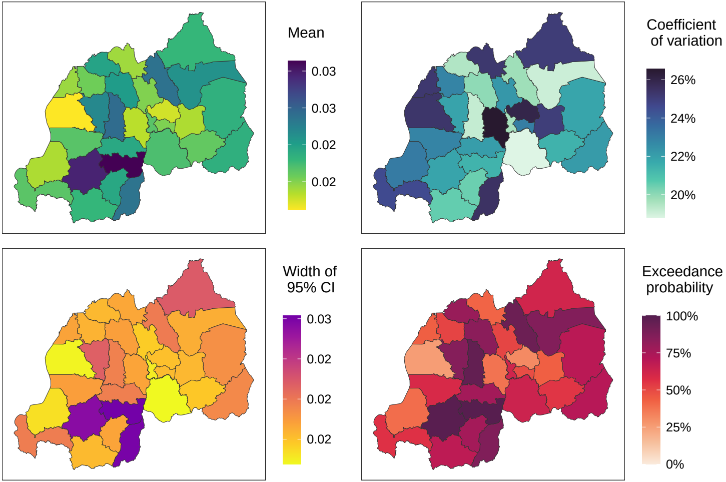

FIGURE 8: Summary measures for area-level (Fay-Herriot) estimates at Admin2.

- Compared to direct estimates (Figure 7) a narrower range in the point estimates.

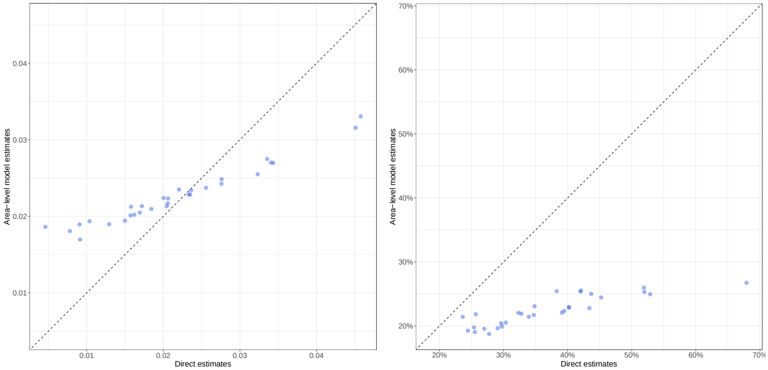

FIGURE 9: Area-level estimates versus direct estimates (left) and area-level CVs versus direct CVs (right), at Admin2.

- As compared to Figure 6, much greater shrinkage at Admin2 (left).
- As compared to Figure 6, much greater gains in precision in all Admin2 areas (right).

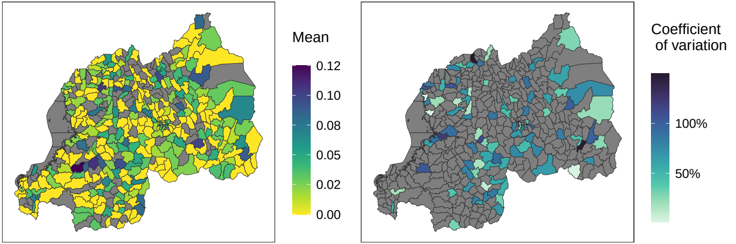

FIGURE 10: Direct prevalence and CV maps at Admin3.

- Direct estimates are not available in quite a few areas (since no data).
- CVs are available in very few, because even if data are available, some configurations do not allow calculation of a variance, e.g., all responses zero.
- So unable to fit area-level models at Admin3 – unit-level models are the only possibility.
- The shinyApp gives warnings.

Data Sparsity Check

	National	Admin-1	Admin-2	Admin-3
Direct Estimates	✔Model is ready to be implemented.	✔Model is ready to be implemented.	✔Model is ready to be implemented.	⚠ Data is not present/sparse in 77% of areas.
Area-level Model		✔Model is ready to be implemented.	✔Model is ready to be implemented.	✘ Data is not present/sparse in 77% of areas, so model will not be fitted.
Unit-level Model		✔Model is ready to be implemented.	✔Model is ready to be implemented.	✔Model is ready to be implemented.

Model Fitting

Run models that passed check

Run all selected models

	National	Admin-1	Admin-2	Admin-3
Direct Estimates	✔ Successful	✔ Successful	✔ Successful	⚠ Model fitted, but interpret with caution due to data sparsity.
Area-level Model		✔ Successful	✔ Successful	Model will not be fitted.
Unit-level Model		✔ Successful	✔ Successful	✔ Successful

FIGURE 11: The shinyApp gives a warning that direct estimates not available in many Admin3 areas, and so this does not allow the fitting of an Admin3 area-level Fay-Herriot model.

Fay-Herriot Modeling

- The F-H approach is an extremely popular way of producing area-level estimates.
- This formulation avoids the need to model the design.
- For a continuous outcome (e.g., child growth measures) it may suffice to simply take the weighted estimator, with its associated variance.
- For prevalences, we can take $g(z) = \log(z/(1-z))$, and then use the delta method to obtain the appropriate variance.
- For rates, we can take $g(z) = \log(z + c)$, where the constant c is chosen to avoid taking the log of zero, and again use the delta method to obtain the appropriate variance.
- Area-level covariates are frequently used – one way of viewing a F-H model is a synthetic estimator with the area-specific error being estimated via a random effects model.
- Problems can occur when the estimate lies on its boundary, both for point estimation and variance estimation (e.g., for a prevalence when all the outcomes are 0).

Inference for the Fay-Herriot Model

Frequentist inference proceeds by integrating out the random effects to give

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}) = \prod_{i=1}^{m} \int_{\delta_i} p(y_i | \delta_i, \boldsymbol{\beta}, \boldsymbol{\theta}) \times p(\delta_i | \boldsymbol{\theta}) \, d\delta_i,$$

where $\boldsymbol{\theta}$ are the variance parameters.

This likelihood may be maximized (using ML or REML) and random effects estimates obtained, via empirical Bayes (EB).

If $\boldsymbol{\beta}, \boldsymbol{\theta}$ known this would produce Best Linear Unbiased Predictors (BLUPs), $\mathrm{E}[\delta_i | \boldsymbol{y}, \boldsymbol{\beta}, \boldsymbol{\theta}]$, with estimates plugged in these become estimated BLUPs (EBLUPs), $\mathrm{E}[\delta_i | \boldsymbol{y}, \widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\theta}}]$.

Inference for the Fay-Herriot Model

In a regular (fixed effects) model, frequentist expectation $\mathrm{E}[\widehat{\boldsymbol{\beta}}]$ and variance $\mathrm{var}(\widehat{\boldsymbol{\beta}})$, are unambiguous.

The frequentist concept of unbiased estimation, and variance estimation, is different in the presence of random effects δ_i.

Unbiased estimation is replaced by (unbiased predictor):

$$\mathrm{E}[\widehat{\delta}_i - \delta_i] = 0,$$

since both $\widehat{\delta}_i$ and δ_i are random variables.

Similarly, the variance is,

$$\mathrm{var}(\widehat{\delta}_i - \delta_i) = \mathrm{MSE}(\widehat{\delta}_i).$$

Asymptotic 95% CIs follow from $\widehat{\delta}_i - \delta_i \sim \mathrm{N}(0, \mathrm{var}(\widehat{\delta}_i - \delta_i))$, are

$$\widehat{\delta}_i \pm 1.95\sqrt{\mathrm{MSE}(\widehat{\delta}_i)}.$$

Frequentist SAE researchers use the mean squared error (MSE) of estimators/predictors as a measure of unceraínty, see for example Datta *et al.* (2011).

The SAR Model

The SAR Model

An alternative to the ICAR model is the simultaneous autoregressive (SAR) Fay-Herriot model.

Following Marhuenda *et al.* (2013), in the `sae` package, a is available, as an alternative to the ICAR model (which is in `SUMMER` and `surveyPrev`).

The SAR model is given by,

$$y_i = \boldsymbol{x}_i^{\mathsf{T}}\beta + \sum_{j=1}^{n} B_{ij}(y_j - \boldsymbol{x}_j^{\mathsf{T}}\beta) + \epsilon_i,$$

for $i = 1, \dots, n$, where

- the SAR coefficients B_{ij} are such that $B_{ii} = 0$,
- ϵ_i are independent, zero mean errors terms with variance V_i,
- Let $\text{var}(\epsilon) = \boldsymbol{V}$ be the $n \times n$ diagonal matrix with i-th element V_i.

In vector form:

$$\boldsymbol{y} - \boldsymbol{x}^\intercal\beta = \boldsymbol{B}(\boldsymbol{y} - \boldsymbol{x}^\intercal\beta) + \epsilon,$$

or

$$(\mathrm{I}_n - \boldsymbol{B})(\boldsymbol{y} - \boldsymbol{x}^\intercal\beta) = \epsilon,$$

or

$$\boldsymbol{y} = \boldsymbol{x}^\intercal\beta + (\mathrm{I}_n - \boldsymbol{B})^{-1}\epsilon,$$

where the latter assumes that $\boldsymbol{B}$ has been chosen so that $\mathrm{I}_n - \boldsymbol{B}$ is invertible.

If $\boldsymbol{B} = \mathbf{0}$ we have an iid model.

So spatial dependence is induced through $(\mathrm{I}_n - \boldsymbol{B})^{-1}$.

Under normality of the errors

$$\boldsymbol{Y} \sim \mathrm{N}(\boldsymbol{x}^\intercal \beta, [(\mathrm{I}_n - \boldsymbol{B})^\intercal \boldsymbol{V} (\mathrm{I}_n - \boldsymbol{B})^{-1}]^\intercal).$$

For a well-defined model we require $(\mathrm{I}_n - \boldsymbol{B})$ to be non-singular, which puts conditions on $\boldsymbol{B}$.

A common choice is

$$\boldsymbol{B} = \rho_s \boldsymbol{W},$$

for a spatial proximity matrix $\boldsymbol{W}$ with elements 1 for neighbors and 0 otherwise.

SAR Models: Some Technical Details

Let $\lambda_{(1)}, \ldots, \lambda_{(n)}$ be the ordered eigenvalues of $\boldsymbol{W}$.

Then, $(\mathrm{I}_n - \boldsymbol{B})$ will be invertible if

$$\frac{1}{\lambda_{(1)}} < \rho_s < \frac{1}{\lambda_{(n)}}.$$

If the row sums of $\boldsymbol{W}$ are standardized to 1 then $\lambda_{(n)} = 1$ and $\lambda_{(1)} \leq -1$ so $\rho_s < 1$ but may be less than -1.

For more discussion, see Banerjee et al. (2015, Section 4.4) and Schabenberger and Gotway (2005, Section 6.2.2.1).

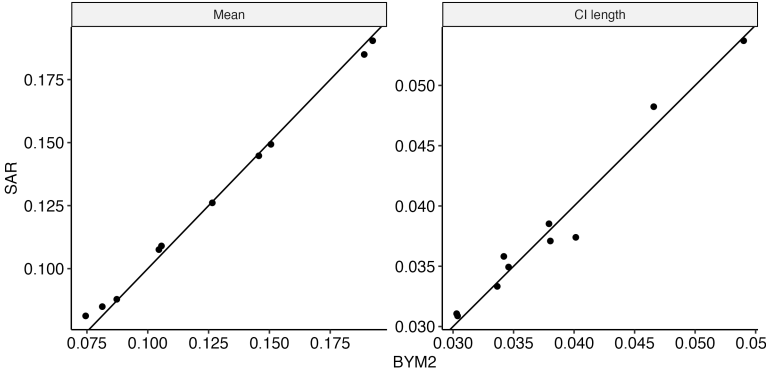

FIGURE 12: Comparison of posterior mean (left column) and 95% credible interval length (right column) under Fay-Herriot models with SAR and BYM2 spatial random effects at Admin1.

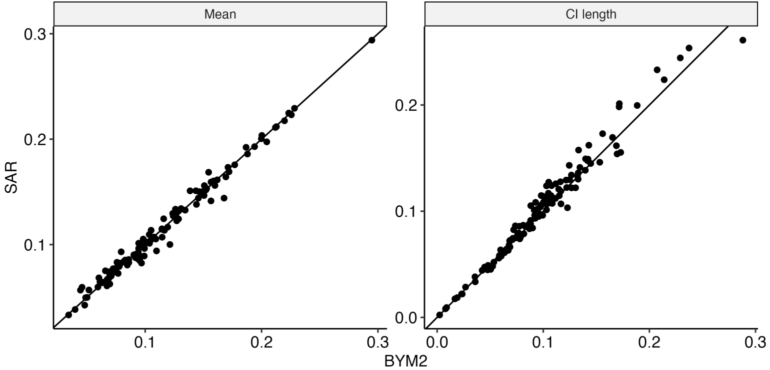

FIGURE 13: Comparison of posterior mean (left column) and 95% credible interval length (right column) under Fay-Herriot models with SAR and BYM2 spatial random effects at Admin2.

Variance Modeling

- When the first-stage sampling variance V_i is unstable, or unavailable in some areas, modeling may be carried out, see for example Otto and Bell (1995).
- Wakefield

Some Variations on F-H Models

- The use of a nonlinear transform at the first stage can give finite sample bias (design-based bias).
- As an alternative, an an unmatched model (You and Rao, 2002) has $z_i = \hat{p}_i^{ht} \sim_{iid} \mathrm{N}(p_i, V_i)$ but with linking model

$$\mathrm{logit}(\mathrm{E}[p_i]) = \alpha + \boldsymbol{x}_i^{\intercal}\boldsymbol{\beta} + \delta_i,$$

in order to avoid small-sample design bias due to the logit transform.

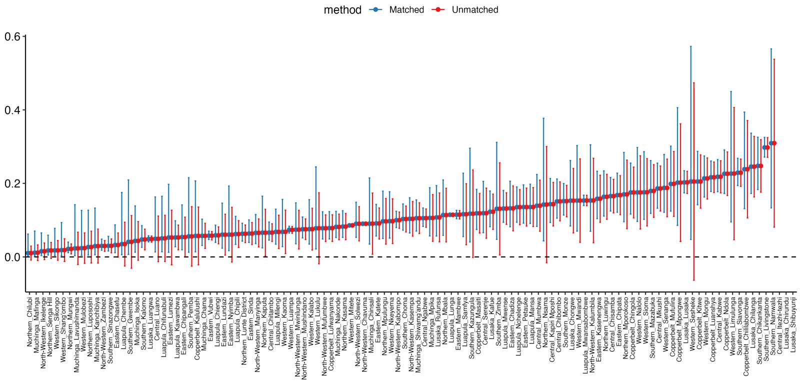

FIGURE 14: Assumed asymptotic sampling distributions, at Admin-2, for unmatched and matched linking models.

Some Variations on F-H Models

Many extensions to the basic F-H model are described in Rao and Molina (2015), including:

- Multivariate models.
- Space-time models (see shortly...).
- Two-fold subarea level models that have random effects for both (say) Admin1 and Admin2 areas.

Software

Software

Discussion

- Fay-Herriot area-level models are by far the most popular SAE models.
- Overshrinkage a worry: nested spatial models are a worthwhile alternative.
- Model checking still in infancy: we use cross-validation to predict weighted estimates.
- Machine learning approaches are very intoxicating, but can the uncertainty be quantified?
- Benchmarking to known totals is less popular in LMICs because the population information may be inaccurate of out of data, see (**?**) for references to the benchmarking literature.
- Fay-Herriot variance modeling can be carried out to increase the utility of this method (Gao and Wakefield, 2023) – a method for variance modification is available in the `surveyPrev` package.

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